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Immune checkpoint inhibitor therapy and risk of type 1 diabetes mellitus in metastatic cancer patients

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Abstract

Aims To assess the risk of new-onset type 1 diabetes mellitus (T1DM) and diabetic ketoacidosis (DKA) in metastatic cancer patients treated with immune checkpoint inhibitors (ICIs) compared to those receiving non-ICI therapies.

Method A retrospective cohort study using TriNetX global electronic health records (2014–2025) from multiple healthcare systems. Adult metastatic cancer patients initiating ICI or non-ICI therapy were included. Patients with preexisting diabetes within 6 months were excluded. After 1:1 propensity score matching, 25,463 patients remained in each group. Outcomes were identified by ICD-10 codes.

Results Median follow-up was 764 days (ICI) vs. 692 days (non-ICI). ICI use was associated with a higher risk of T1DM (HR, 2.35; 95% CI, 1.81–3.04) and DKA (HR, 10.58; 95% CI, 4.21–26.59). Cumulative incidence analyses supported these findings, with ICIs showing higher risks of T1DM (0.75% vs. 0.32%; RR, 2.32 [95% CI, 1.79–3.00]) and DKA (0.20% vs. 0.04%; RR, 5.00 [95% CI, 2.54–9.86]). Subgroup analyses identified elevated baseline HbA1c (> 6.0%), male sex, white race, and dual checkpoint blockade as high-risk factors.

Conclusion ICIs significantly increase the risk of T1DM and DKA. These findings highlight the need for vigilant glycemic monitoring in cancer patients treated with ICIs, especially within identified high-risk subgroups.

Keywords Immune checkpoint inhibitors, Type 1 diabetes, Diabetic ketoacidosis, Cancer immunotherapy, Immune-Related adverse events, Propensity score matching

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Introduction

Immune checkpoint inhibitors (ICIs) have revolutionized cancer treatment by enabling durable tumor control through enhanced anti-tumor T-cell responses [1, 2]). Targeting immune regulators like PD-1, PD-L1, or CTLA-4, ICIs significantly improve survival in melanoma, lung cancer, and renal cell carcinoma [3–5]. However, disrupting immune checkpoints can also trigger immune-related adverse events (irAEs) [6, 7].

Endocrine irAEs, including thyroiditis, hypophysitis, and insulin-dependent diabetes mellitus resembling type 1 diabetes (ICI-T1DM), are particularly significant [8, 9]. Though rare (under 1%), ICI-T1DM is often abrupt, permanent, and life-threatening, frequently presenting as diabetic ketoacidosis (DKA) [10–13]. Typically, patients display extreme hyperglycemia, ketosis, and negligible C-peptide levels, indicating extensive β -cell destruction. Islet autoantibodies are variably present, distinguishing it from typical type 1 diabetes [14–16].

The precise mechanisms of ICI-T1DM remain under investigation. Normally, PD-1 and CTLA-4 pathways maintain immune tolerance and prevent autoimmunity [17, 18]. ICIs disrupt cancer cell immune evasion by promoting T-cell activation, potentially activating autoreactive CD8⁺ cytotoxic lymphocytes targeting pancreatic β -cells [19, 20]. [21] Specific genetic factors, notably certain HLA haplotypes, also increase vulnerability [22]. Unlike classic T1DM, islet autoantibodies are less consistently found in ICI-T1DM, although when present, diabetes onset typically occurs earlier [21–23].

Real-world incidence data and risk factors remain poorly defined. Current evidence mainly comprises case reports or clinical trials without comparison groups [24, 25]. Chen et al. reported a 0.86% incidence in ICI-treated patients, highest with combined PD-1 and CTLA-4 inhibitors and pre-existing type 2 diabetes, but lacked comparison to non-ICI-treated groups [10].

This study used a large electronic health database to compare diabetes risk in ICI-treated cancer patients against those treated with non-ICI therapies, aiming to quantify the excess risk from ICIs, identify associated patient or treatment factors, and characterize clinical presentation and timing. We hypothesized that ICIs increase risks of acute autoimmune diabetes and DKA even after adjusting for baseline factors, particularly in certain high-risk subgroups.

Methods

Study design and data source We performed a retrospective cohort study to compare diabetes-related outcomes in metastatic cancer patients initiating ICI therapy versus those initiating non-immune cancer therapy. Data were obtained from the TriNetX global health research network, a federated electronic health records reposi-

tory containing de-identified patient data from multiple healthcare systems worldwide. TriNetX aggregates clinical information directly from electronic medical records of over 200 healthcare organizations across 30 countries, providing access to longitudinal data from more than 200 million patients. The platform harmonizes data to common terminologies including ICD-10-CM for diagnoses, CPT and ICD-10-PCS for procedures, LOINC for laboratory results, and RxNorm for medications. All data within TriNetX are de-identified in accordance with HIPAA and GDPR standards. This database offers a comprehensive view of real-world clinical practice, enabling large-scale comparative effectiveness research across diverse patient populations. The study period spanned patients treated from 2014 to 2025. The study was reviewed and approved by the institutional review board with a waiver of informed consent due to use of de-identified data (TSGHIRB No. E202516013).

Cohort Selection: We identified an ICI-treated cohort of adult cancer patients who received at least one dose of an FDA-approved immune checkpoint inhibitor (e.g. anti-PD-1, anti-PD-L1, or anti-CTLA-4 agents) during the study period. The comparison cohort consisted of adult cancer patients treated with non-ICI systemic therapy, defined as either conventional cytotoxic chemotherapy or therapeutic monoclonal antibodies not targeting immune checkpoints. To ensure new-onset outcomes, we required patients to have no recorded diagnosis of diabetes (of any type) or DKA prior to the start of therapy (index date). We also excluded patients with less than 6 months of follow-up post-treatment initiation, to allow adequate observation time for outcome development. Patients in the ICI cohort were matched 1:1 to patients in the non-ICI cohort using propensity scores to control for confounding factors (Supplement Fig. 1).

Propensity Score Matching: A propensity score for receiving ICI (versus non-ICI therapy) was calculated for each patient using a multivariable logistic regression model. Covariates in this model included: age, sex, race/ethnicity, baseline body mass index (BMI), baseline hemoglobin A1c (HbA1c), and relevant comorbidities such as hypertension, hyperlipidemia, coronary artery disease, heart failure, and chronic lung disease. We also incorporated cancer type and stage (when available), to account for disease severity, since certain ICIs are preferentially used in advanced disease. Patients were matched on the logit of the propensity score using nearest-neighbor matching without replacement (caliper=0.2 of the standard deviation of the logit). Matching quality was assessed by standardized mean differences (SMD) for each covariate before and after matching, with SMD<0.1 indicating good balance. After matching, the cohorts were examined to ensure no substantial residual differences in baseline characteristics.

Exposure definition The exposure of interest was initiation of ICI therapy. ICIs included FDA-approved agents targeting PD-1 (nivolumab, pembrolizumab, cemiplimab), PD-L1 (atezolizumab, durvalumab, avelumab), or CTLA-4 (ipilimumab, tremelimumab). Combination ICI therapy was defined as overlapping treatment with two classes of ICIs (e.g., concurrent PD-1 and CTLA-4 blockade). The comparison exposure was non-immune cancer therapy, which included traditional chemotherapy agents and non-ICI targeted therapies (such as VEGF or EGFR inhibitors, and other monoclonal antibodies not directed at T-cell checkpoints). Patients remained in their initial exposure group for analysis (an “as-treated” analysis by initial assignment).

Outcome measures We focused on diabetes-related outcomes occurring after treatment initiation. The primary outcomes were new-onset type 1 diabetes mellitus (T1DM) and diabetic ketoacidosis (DKA). Secondary outcomes included new-onset type 2 diabetes mellitus (T2DM), hyperglycemia (impairment of glucose tolerance not yet classified as diabetes, ICD-10 R73.9), hyperosmolar hyperglycemic state (HHS), and composite diabetes-related complications (a combined endpoint including diabetic nephropathy, retinopathy, or neuropathy). Outcomes were identified by the presence of newly assigned ICD-10 diagnosis codes in the post-index period. To enhance specificity, we required at least two separate outpatient or problem-list codes for a new diagnosis of T1DM or T2DM (to exclude transient hyperglycemia or coding error), and at least one code for acute conditions like DKA, HHS, or hyperglycemia. “Type 1 DKA” was defined as DKA occurring in a patient coded as T1DM; “Type 2 DKA” was DKA in a patient coded as T2DM, acknowledging that ketosis can occur in severe T2DM as well. All detailed definitions of the primary outcomes was shown in Supplement Table 1.

Statistical analysis: We compared outcome incidence between the matched ICI and non-ICI cohorts. Time-to-event analyses were performed using Kaplan-Meier curves and Cox proportional hazards models to calculate hazard ratios (HRs) for developing each outcome, with censoring at last follow-up. In addition, cumulative incidence was estimated, and relative risks (RRs) with 95% confidence intervals were calculated for each outcome as supportive analyses. We conducted stratified subgroup analyses to identify effect modifiers: subgroups by age (18–45, 45–64, 65–79, and ≥ 80 years), sex, race (White, Black and Asian), obesity (BMI ≥ 30 vs. < 30), baseline glycemic status (HbA1c $> 6.0\%$ vs. $\leq 6.0\%$), cancer subtype (Melanoma, lung, genitourinary, digestive and breast) and therapy type (single-agent ICI vs. dual ICI combination). The significance of interaction terms between treatment and subgroup factors was tested in the models.

We also performed sensitivity analyses to test robustness of results: (1) restricting to patients with at least 12 documented treatment administration records within 6 months (to ensure adequate treatment exposure and follow-up, i.e. an “on-treatment” analysis); and (2) restricting to patients from the United States only (to account for any international differences in coding or practice). Finally, we examined negative control outcomes — health outcomes not plausibly related to immune therapy (such as incidence of accidental injuries or unrelated conditions) — to detect any spurious associations due to residual confounding. To address geographic heterogeneity, we included region indicators (United States, EMEA, Asia-Pacific, Latin America) in adjusted models where available and conducted region-stratified analyses. All analyses were conducted using SAS and R software. Statistical significance was defined as $p < 0.05$ (two-tailed).

Results

Baseline characteristics

From an initial pool of 27,126 ICI-treated and 204,597 non-ICI-treated cancer patients, propensity matching yielded 25,463 patients per group with well-balanced baseline characteristics (Table 1). The matched cohorts had an average age of 63 years (SD ~ 12.7 years), were predominantly male (52%), and included diverse racial/ethnic backgrounds (68% White, 7% Black, 7% Asian, 14% unknown). Comorbidities such as hypertension (33%), hyperlipidemia (21%), median BMI (27.2 kg/m²), and median HbA1c (5.8%) were evenly distributed. Cancer types differed slightly; ICI patients frequently had thoracic, skin, and urinary cancers, whereas digestive and breast cancers were more common among non-ICI patients, though matching accounted for cancer type.

Time-to-event analysis of diabetes-related outcomes: During a median follow-up of 764 days (interquartile range: 1111 days) in the ICI cohort and 692 days (interquartile range: 1257 days), Kaplan-Meier analysis demonstrated that ICI treatment significantly increased diabetes-related risks (Supplement Fig. 2). The hazard ratio (HR) for new-onset type 1 diabetes mellitus (T1DM) was 2.35 (95% CI, 1.81–3.04; $p < 0.001$). Strikingly, diabetic ketoacidosis (DKA) had an HR of 10.58 (95% CI, 4.21–26.59), highlighting substantial early risk following ICI initiation. Elevated risks extended to type 2 diabetes (T2DM, HR 1.40, 95% CI, 1.31–1.49), T2DM-related DKA (HR 5.10, 95% CI, 2.93–8.88), and hyperosmolar hyperglycemic state (HR 1.67, 95% CI, 1.09–2.55), indicating a broad metabolic impact of ICIs (Fig. 1).

Cumulative incidence of new-onset diabetes mellitus Consistent with time-to-event analyses, cumulative incidence revealed notably higher rates of T1DM (0.75% vs. 0.32%; RR 2.32, 95% CI, 1.79–3.00, $p < 0.001$) and T2DM

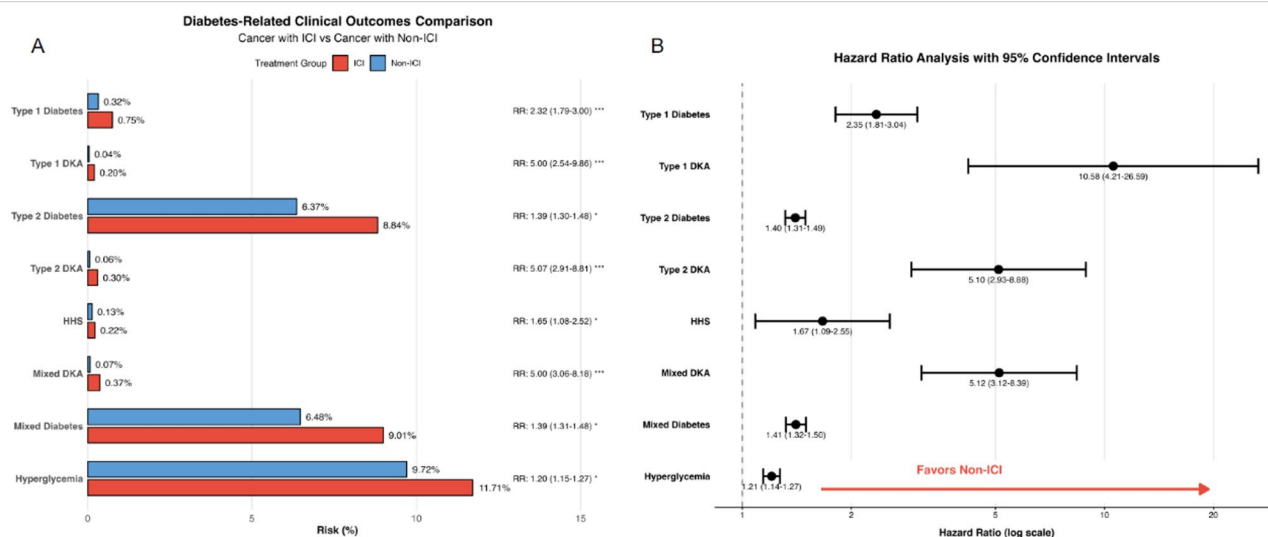


Fig. 1 Diabetes-Related Clinical Outcomes Comparison and Hazard Ratio Analysis. **PanelA:** Risk Comparison Between Treatment Groups. Bar chart showing the incidence (%) of diabetes-related outcomes in cancer patients treated with immune checkpoint inhibitors (red) versus conventional chemotherapy (blue). Risk ratios with 95% confidence intervals and significance levels are displayed for each outcome. ** indicates $P < 0.01$. **PanelB:** Hazard Ratio Analysis with 95% Confidence Intervals. Forest plot showing the hazard ratios for diabetes-related outcomes comparing immune checkpoint inhibitor therapy to conventional chemotherapy. Hazard ratios are displayed on a logarithmic scale, with values > 1 indicating higher risk with immune checkpoint inhibitor therapy

(8.84% vs. 6.37%; RR 1.39, 95% CI, 1.30–1.48, $p < 0.001$) among ICI patients (Fig. 1). Hyperglycemia without diabetes diagnosis also modestly increased (11.71% vs. 9.72%; RR 1.20, 95% CI, 1.15–1.27).

Diabetic ketoacidosis and severe hyperglycemic events DKA was a relatively rare outcome but demonstrated the most pronounced difference between groups. In total, there were very few DKA events in the non-ICI cohort ($< 0.1\%$ incidence), whereas the ICI cohort had significantly more. Among cases coded as T1DM, DKA occurred in 0.20% of patients on ICIs versus 0.04% on non-ICI therapy, translating to RR 5.00 (95% CI 2.54–9.86) — a five-fold higher risk of life-threatening ketoacidosis attributable to ICI treatment (Fig. 1). This risk ratio, while substantial, was lower than the corresponding hazard ratio (10.58), suggesting that the instantaneous risk of DKA is particularly high early in the course of ICI treatment.

DKA episodes in patients coded as T2DM (which may represent ketosis-prone type 2 diabetes or miscoded type 1) were also more frequent with ICIs (0.30% vs. 0.06%, RR 5.07, 95% CI 2.91–8.81). Similarly, hyperosmolar hyperglycemic state was reported infrequently, but more often in the ICI group (incidence 0.22% vs. 0.13%, RR 1.65, 95% CI 1.08–2.52). Figure 2 illustrates the cumulative incidence of new-onset T1DM and DKA over time by treatment group: the curves for the ICI cohort start to diverge early in follow-up and continue to rise above the non-ICI cohort over several years.

We observed that 50% of the ICI-T1DM cases had onset within approximately 2–3 years of starting ICI. Indeed, an early-onset pattern was evident: by 500 days into therapy, the incidence of new T1DM in the ICI group was already significantly higher than in controls (with about 0.3% of ICI patients developing T1DM by 500 days vs. $< 0.1\%$ of controls).

Subgroup analyses Risks varied across subgroups (Figs. 3 and 4). Elevated baseline HbA1c ($> 6.0\%$) significantly increased the risk of T1DM (HR 2.48 compared to 1.54 with normal HbA1c) and T1DM-associated ketoacidosis (HR 9.27 compared to 2.15 with normal HbA1c). Male patients demonstrated greater susceptibility to both T1DM and T1DM-related ketoacidosis. White patients exhibited higher relative risks for T1DM and T1DM-associated ketoacidosis compared with Black and Asian patients. Results were consistent across age categories, and risks of T1DM and DKA were most pronounced in melanoma. However, no notable risk difference was observed based on BMI. Additionally, dual checkpoint inhibitor therapy substantially amplified the risks of T1DM (HR 3.10) and diabetic ketoacidosis (HR 5.11).

Sensitivity analyses Results were robust across sensitivity analyses (Supplement Table 5), which confirmed robustness of findings, showing similar or increased risks in subsets with extended treatment and U.S.-only patients. Negative control outcomes demonstrated no significant

Table 1 Baseline characteristics of patients treated with immune therapy vs. Non-Immune therapy

Characteristics	Before Propensity Score Matching				After Propensity Score Matching			
	ICI (n = 27,126)	Non-ICI (n = 204,597)	P Value	SMDs	ICI (n = 25,463)	Non-ICI (n = 25,463)	P Value	SMDs
Demographics								
Age at index, mean (SD), y	63.0 (12.7)	59.9 (13.9)	< 0.001	0.231	63.0 (12.7)	63.0 (12.9)	0.518	0.006
Sex								
Female, No. (%)	11,009 (40.5%)	103,807 (50.7%)	< 0.001	0.258	11,008 (43.2%)	10,898 (42.8%)	0.325	0.009
Male, No. (%)	13,177 (48.5%)	76,608 (37.4%)	< 0.001	0.21	13,173 (51.7%)	13,347 (52.4%)	0.123	0.014
Unknown Sex	2,940 (10.8%)	24,182 (11.8%)	0.888	0.031	1,282 (5.0%)	1218 (4.8%)	0.189	0.012
Race/ethnicity								
White, No. (%)	17,270 (67.8%)	103,928 (56.1%)	< 0.001	0.244	17,267 (67.8%)	17,337 (68.1%)	0.506	0.006
Black or African American, No. (%)	1,748 (6.9%)	13,931 (7.5%)	< 0.001	0.025	1,748 (6.9%)	1,741 (6.8%)	0.902	0.001
Asian, No. (%)	1,798 (7.1%)	24,007 (13.0%)	< 0.001	0.197	1,798 (7.1%)	1,814 (7.1%)	0.782	0.002
Unknown Race, No. (%)	6,310 (23.2%)	62,731 (30.6%)	< 0.001	0.167	4,650 (18.2%)	4,571 (17.9%)	0.363	0.008
Comorbidities								
Essential hypertension, No. (%)	8,567 (33.6%)	48,890 (26.4%)	< 0.001	0.159	8,563 (33.6%)	8,460 (33.2%)	0.333	0.009
Hyperlipidemia, No. (%)	5,360 (21.0%)	26,556 (14.3%)	< 0.001	0.177	5,355 (21.0%)	5,281 (20.7%)	0.42	0.007
Heart failure, No. (%)	1,005 (3.9%)	5,580 (3.0%)	< 0.001	0.051	1,004 (3.9%)	875 (3.4%)	0.002	0.027
Acute myocardial infarction, No. (%)	569 (2.2%)	2,676 (1.4%)	< 0.001	0.059	569 (2.2%)	545 (2.1%)	0.467	0.006
Cerebral infarction, No. (%)	559 (2.2%)	2,743 (1.5%)	< 0.001	0.053	557 (2.2%)	513 (2.0%)	0.174	0.012
Clinical measurements								
Malignant neoplasms of respiratory and intrathoracic organs	5,292 (20.8%)	20,799 (11.2%)	< 0.001	0.263	5,290 (20.8%)	5,215 (20.5%)	0.411	0.007
Melanoma and other malignant neoplasms of skin	4,175 (16.4%)	8,667 (4.7%)	< 0.001	0.389	4,170 (16.4%)	4,038 (15.9%)	0.112	0.014
Malignant neoplasms of digestive organs	2,652 (10.4%)	39,121 (21.1%)	< 0.001	0.297	2,652 (10.4%)	2,735 (10.7%)	0.232	0.011
Malignant neoplasms of urinary tract	2,686 (10.5%)	6,232 (3.4%)	< 0.001	0.285	2,681 (10.5%)	2,684 (10.5%)	0.965	< 0.001
Neoplasms of uncertain behavior, polycythemia vera and myelodysplastic syndromes	2,579 (10.1%)	14,982 (8.1%)	< 0.001	0.071	2,578 (10.1%)	2,452 (9.6%)	0.061	0.017
Malignant neoplasms of female genital organs	1,724 (6.8%)	13,012 (7.0%)	0.14	0.011	1,724 (6.8%)	1,802 (7.1%)	0.173	0.012
Malignant neoplasms of lip, oral cavity and pharynx	1,324 (5.2%)	7,671 (4.1%)	< 0.001	0.05	1,324 (5.2%)	1,415 (5.6%)	0.074	0.016
Malignant neoplasms of breast	1,245 (4.9%)	32,053 (17.3%)	< 0.001	0.403	1,245 (4.9%)	1,198 (4.7%)	0.33	0.009
Malignant neoplasms of male genital organs	994 (3.9%)	10,257 (5.5%)	< 0.001	0.077	994 (3.9%)	944 (3.7%)	0.247	0.01
Malignant neoplasms of lymphoid, hematopoietic and related tissue	699 (2.7%)	9,349 (5.0%)	< 0.001	0.119	699 (2.7%)	607 (2.4%)	0.01	0.023
Clinical measurements								
BMI, mean (SD)	27.2 (6.2)	27.1 (6.3)	0.207	0.011	27.2 (6.2)	27.3 (6.1)	0.405	0.01
Hemoglobin A1c, %, mean (SD)	5.8 (1.4)	5.7 (1.2)	0.001	0.046	5.8 (1.4)	5.8 (1.4)	0.808	0.005
Cholesterol, mg/dL, mean (IQR)	176.4 (Q1 144.52–Q3 208.28)	182.0 (Q1 149.78–Q3 214.22)	< 0.001	0.118	176.4 (Q1 144.52–Q3 208.28)	179.0 (Q1 146.85–Q3 211.15)	0.001	0.056
Triglyceride, mg/dL, mean (IQR)	121.8 (Q1 72.06–Q3 171.54)	122.3 (Q1 72.22–Q3 172.38)	0.565	0.007	121.8 (Q1 72.06–Q3 171.54)	122.2 (Q1 73.54–Q3 170.86)	0.77	0.005

BMI indicates body mass index; ICI, immune therapy; SD, standard deviation; SMD, standardized mean difference. * Indicates characteristics that remain statistically different ($P < 0.05$) after propensity score matching. SMD < 0.1 indicates balanced characteristics between groups

differences, minimizing concerns of surveillance bias or unmeasured confounding.

Region-stratified analyses showed directionally consistent ICI-associated hazards across the United States, EMEA, and Asia–Pacific, with similar trends and wider CIs in Latin America; region-adjusted models yielded

estimates unchanged within rounding (Supplement Table 6).

Overall, results firmly establish that ICI therapy significantly elevates risks of diabetes mellitus and severe hyperglycemic complications, underscoring the need for proactive monitoring and patient education strategies.

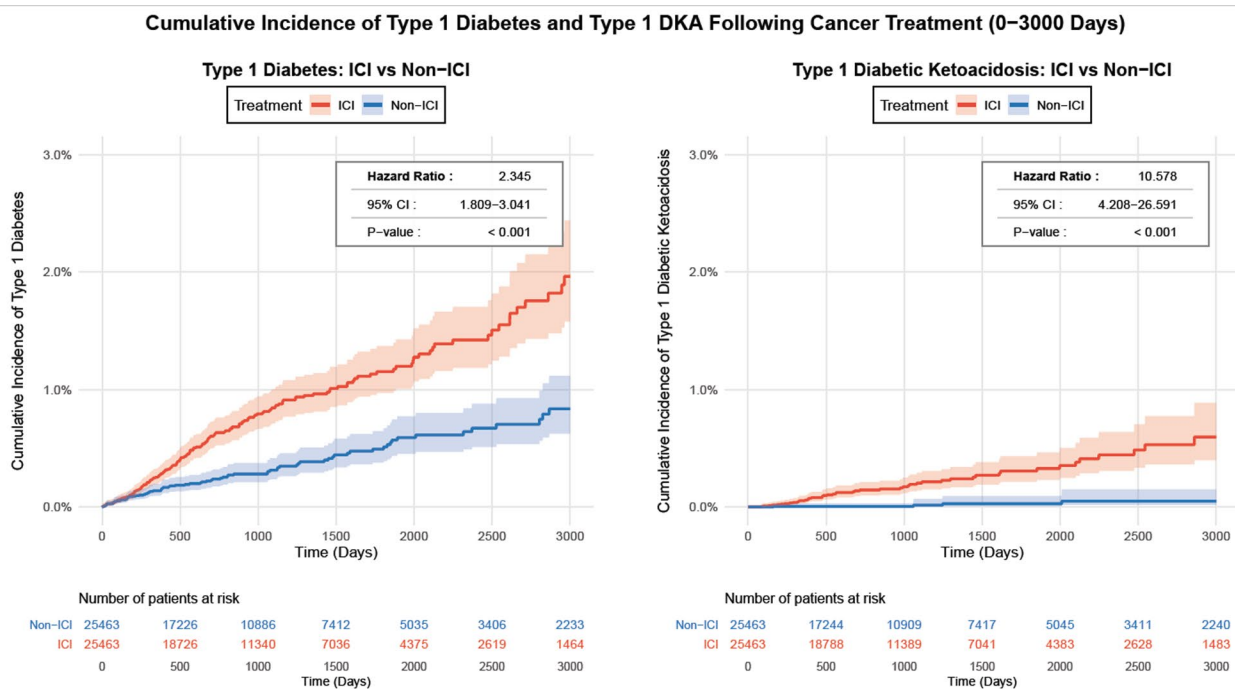


Fig. 2 Cumulative Incidence of Type 1 Diabetes and Type 1 DKA Following Cancer Treatment. Kaplan-Meier curves showing the cumulative incidence of Type 1 Diabetes (left panel) and Type 1 Diabetic Ketoacidosis (right panel) in patients treated with immune therapy (red lines) versus chemotherapy (blue lines). The hazard ratio for Type 1 Diabetes was 2.345 (95% CI: 1.809–3.041, $P < 0.001$) and for Type 1 DKA was 10.578 (95% CI: 4.208–26.591, $P < 0.001$). The number of patients at risk at different time points is shown below each graph

Discussion

In this large real-world cohort study using propensity score matching, we found that cancer patients treated with immune checkpoint inhibitors had a substantially higher risk of developing new-onset insulin-dependent diabetes and diabetic ketoacidosis compared to patients treated with non-ICI treatment. The hazard ratio for developing T1DM was 2.35 (95% CI, 1.81–3.04), and for DKA it was strikingly elevated at 10.58 (95% CI, 4.21–26.59), representing a ten-fold increase in the instantaneous risk of this life-threatening complication. These findings were supported by cumulative incidence analyses showing a 2.32-fold relative risk increase for T1DM and a 5-fold increase for DKA. Although the absolute incidence of ICI-induced T1DM was low (0.75% in our cohort), the magnitude of relative risk increase is clinically significant given the severity of this adverse event.

Our results are consistent with prior findings. For instance, Chen et al. reported a similar incidence (0.86%) of ICI-associated T1DM in a large cohort of approximately 30,000 ICI-treated patients, observing virtually no cases in the non-ICI comparator group [10]. They also noted increased risk with combined PD-1/PD-L1 plus CTLA-4 blockade (HR approximately 1.6 vs. single-agent), aligning with our observed HR of 3.10 for dual checkpoint therapy. Unlike Chen et al.'s study, our analysis explicitly excluded patients with preexisting diabetes,

allowing clear evaluation of de novo autoimmune diabetes risk. Additionally, our longer median follow-up (median 764 days vs. approximately 1 year in most prior studies) provided a more comprehensive assessment of timing and onset patterns.

This study also provides new insights into patient-level risk factors for ICI-T1DM. We observed a strong association between pre-existing metabolic derangements and heightened risk of ICI-induced diabetes. Patients with elevated baseline HbA1c ($> 6.0\%$) had a significantly greater likelihood of developing T1DM (HR 2.48) or DKA (HR 9.27) after starting ICIs compared to normoglycemic patients. Chen et al. similarly reported a 4-fold increase in ICI-T1DM risk in patients with pre-existing (non-T1DM) diabetes [10]. This suggests that patients with underlying β -cell stress or insulin resistance may be more susceptible to the additional autoimmune insult triggered by ICIs.

Male sex was another significant risk factor, with males demonstrating a hazard ratio of 2.79 for T1DM versus 1.78 in females. Additionally, racial disparities emerged, showing white patients having a higher hazard ratio (HR 2.49) compared to Black (HR 1.46) and Asian (HR 1.05) patients. These differences may partly relate to genetic variations, including differential HLA distribution by ethnicity [26, 27]. Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) showed a similar hazard for T1DM (HR 2.50) compared to non-obese patients (HR 2.17). Intriguingly, however, non-obese

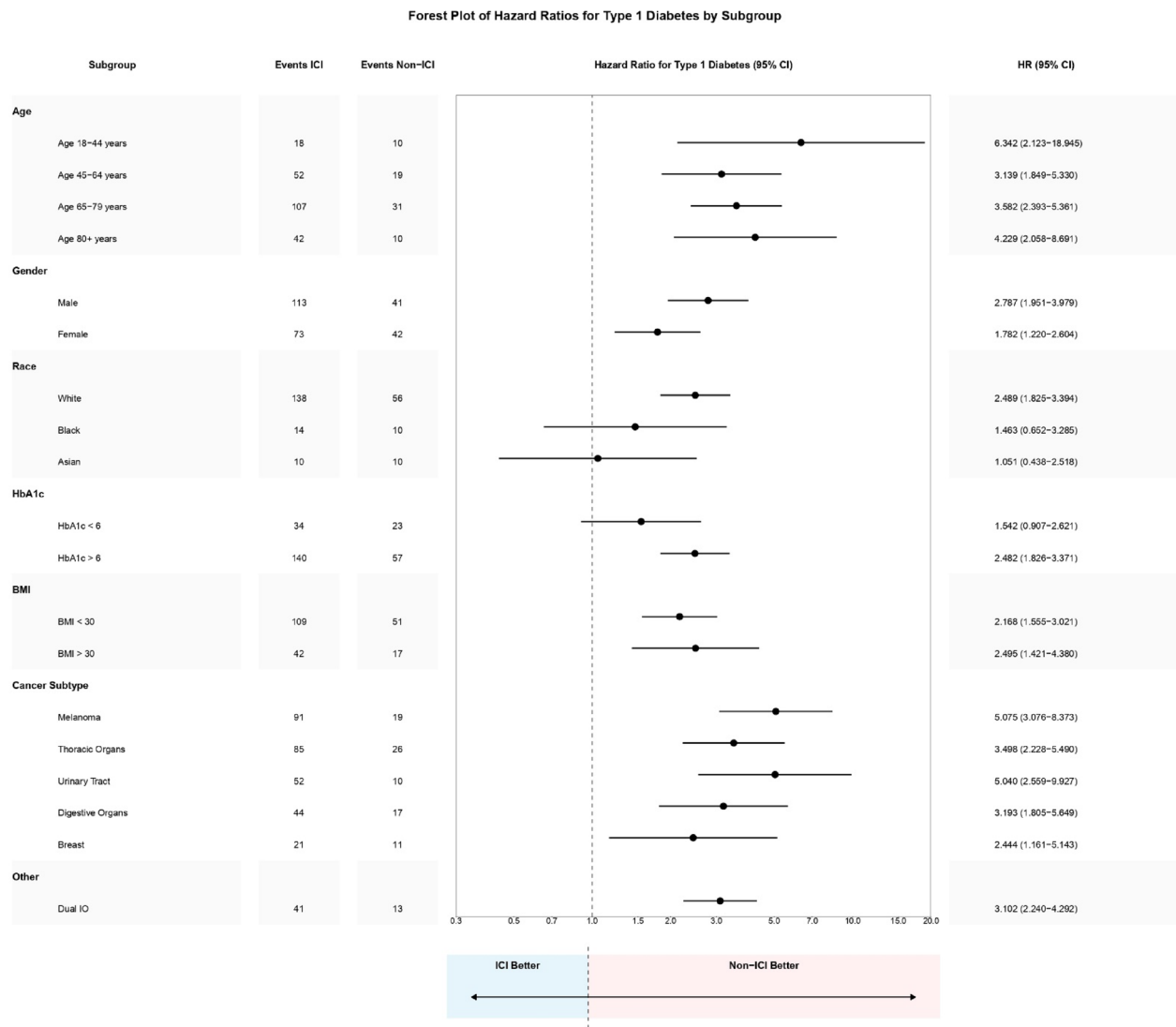


Fig. 3 Subgroup Hazard Ratios for New-Onset T1DM after ICI vs. non-ICI. Hazard ratios with 95% confidence intervals for developing Type 1 Diabetes with immune checkpoint inhibitor therapy versus non-ICI therapy, stratified by demographic and clinical characteristics. Values shown on the right represent hazard ratios with corresponding 95% confidence intervals. The figure demonstrates consistently increased risk across most subgroups, with particularly elevated risk among males, white patients, those with HbA1c ≥6, and patients receiving dual immunotherapy

patients exhibited a notably higher risk for rapid progression to DKA (HR 6.34) compared to obese counterparts (HR 2.13). This observation suggests obesity may influence disease progression rather than risk of initial onset.

Dual checkpoint inhibition emerged as the strongest treatment-related risk factor, with a 3.10-fold hazard of T1DM and 5.11-fold hazard of DKA compared to matched controls. The substantially higher risk with combination therapy versus monotherapy aligns with observations across other immune-related adverse events and supports the hypothesis that more intensive immune activation increases autoimmunity risk. This finding has important implications for risk stratification in patients receiving combination ICI regimens.

The clinical presentation of ICI-induced diabetes in our cohort was typically abrupt and severe, consistent with published literature. The hazard ratio for DKA (10.58) was notably higher than the risk ratio (5.00), suggesting that the risk is particularly concentrated early in treatment. Indeed, we observed that about two-thirds of ICI-T1DM cases arose within the first 6 months of therapy. This early timing coincides with the period of most vigorous immune activation by the checkpoint inhibitor. However, late-onset cases can occur; we observed some patients developing diabetes a year or more into treatment. This implies that patients remain at risk for autoimmune β-cell destruction as long as immune system activation continues.

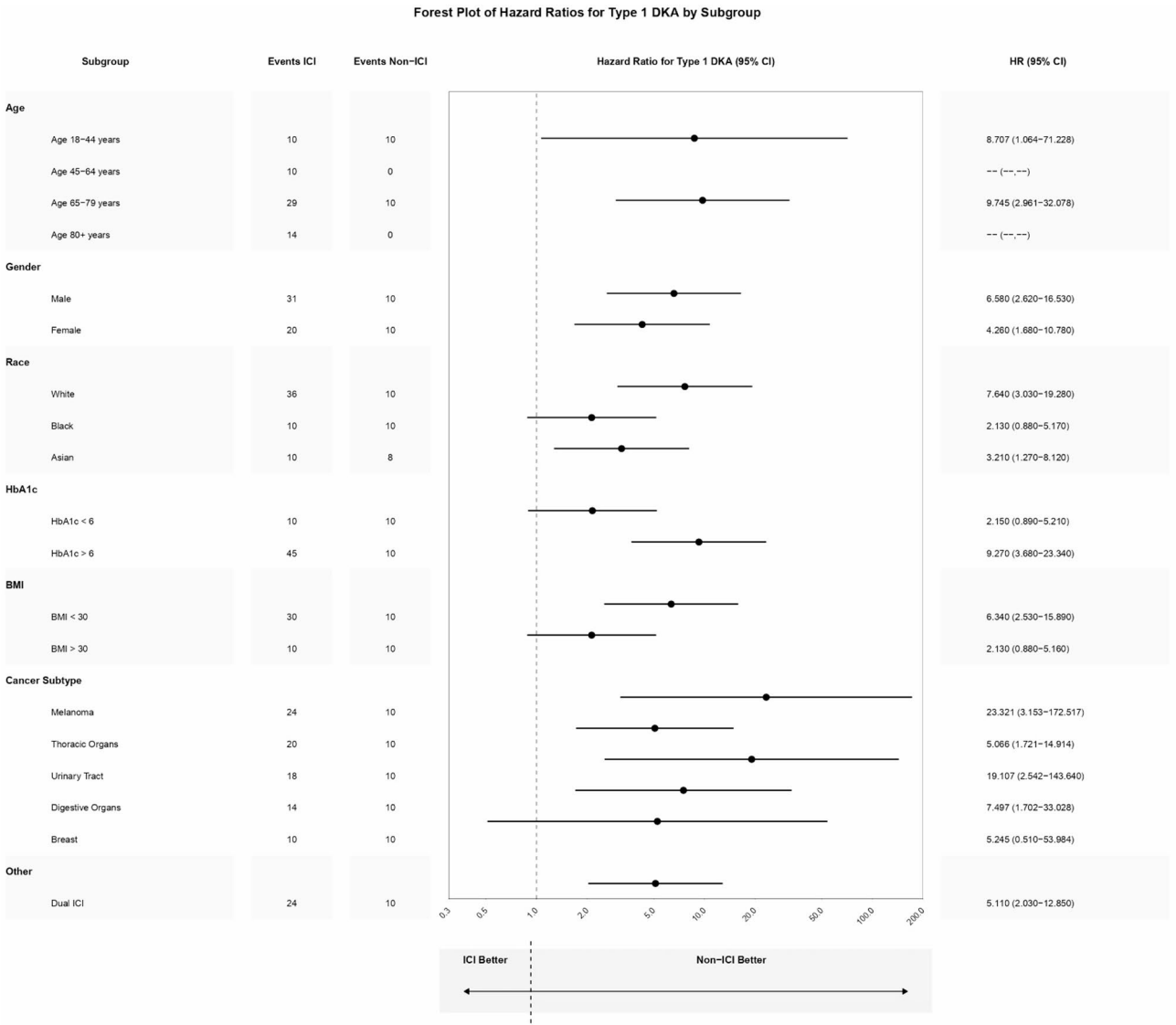


Fig. 4 Subgroup Hazard Ratios for T1DM-associated DKA after ICI vs. non-ICI. Hazard ratios with 95% confidence intervals for developing Type 1 Diabetic Ketoacidosis with immune checkpoint inhibitor therapy versus non-ICI therapy, stratified by demographic and clinical characteristics. The forest plot demonstrates markedly higher risk in most subgroups, with particularly elevated hazard ratios for patients with HbA1c ≥ 6 (HR 9.27), white patients (HR 7.64), and males (HR 6.58)

Our results have several clinical implications. Foremost, they emphasize the importance of vigilant monitoring for hyperglycemia in patients receiving ICIs. Given how sudden ICI-T1DM can be, we recommend checking blood glucose or HbA1c prior to each ICI infusion or at least every few weeks during active therapy, especially for high-risk patients. Our data suggest that individuals with baseline HbA1c > 6.0%, male sex, white race, and those receiving dual checkpoint blockade should be monitored particularly closely. Educate patients and clinicians to recognize early warning signs of new-onset hyperglycemia and impending DKA—mild hyperglycemia, polyuria, polydipsia, nausea/vomiting, abdominal pain, dyspnea, and unexplained fatigue—and to contact the oncology

team promptly or seek urgent/emergency care. Evaluation should include immediate measurement of blood (or capillary) glucose, blood or urine ketones, and, when feasible, serum C-peptide to identify insulin deficiency and the risk of progression to DKA. Educating patients about these symptoms and providing glucose meters for self-monitoring may facilitate earlier detection and management, reducing the risk of severe complications like DKA. When ICI-induced diabetes is confirmed, management should involve a multidisciplinary approach. The cornerstone of treatment is insulin therapy, often requiring a basal-bolus regimen akin to type 1 diabetes management [21]. Unlike some other irAEs, high-dose steroids is not effective in

reversing established ICI-T1DM, presumably because β -cell destruction is rapid and irreversible by the time of diagnosis [23]. Insulin is therefore the only suitable treatment to control blood glucose in these patients.

Our study possesses several important strengths. First, the large real-world sample size drawn from multiple centers across various cancer types enhances generalizability. Utilizing a matched comparator group allowed us to effectively isolate the impact of immune checkpoint inhibitors (ICIs) on diabetes risk. Rigorous control for known confounders, extensive sensitivity analyses, and assessment of negative control outcomes further bolstered the validity of our findings, reducing the likelihood of artifacts. Our analysis provides critical real-world evidence complementary to clinical trial data, where rare events like autoimmune diabetes are frequently underreported. Unlike previous large-scale studies, we specifically excluded patients with preexisting diabetes and provided direct risk comparisons to non-ICI chemotherapy, thus offering precise risk estimates for newly developed diabetes. Additionally, our extended follow-up period (median approximately 2 years vs. 1 year in most prior studies) allowed for better identification and characterization of late-occurring cases.

Nevertheless, our study also has notable limitations. Potential residual confounding remains due to unmeasured factors such as family history, lifestyle behaviors, and undocumented medications. Because country-level identifiers are not uniformly available in TriNetX, we could not adjust by country; however, region adjustment and region-stratified analyses yielded consistent associations across settings (as shown in supplement Table 6), supporting robustness despite international heterogeneity. Our reliance on diagnostic codes may introduce misclassification, although requiring ≥ 2 confirmations for chronic outcomes improved specificity. Detailed clinical biochemistry was limited: islet autoantibody titers, HLA typing, precise timing of ICI relative to diabetes onset, and comprehensive corticosteroid exposure were not uniformly captured. Although TriNetX includes some laboratory fields (e.g., C-peptide, pH, bicarbonate), measurement frequency and time stamps vary by site and are not consistently aligned with acute events; accordingly, we prespecified ICD-10–based definitions for T1DM and DKA to minimize differential misclassification. The low incidence of ICI-induced T1DM restricts event numbers and widens confidence intervals—particularly in subgroup analyses—so results should be interpreted cautiously pending confirmation in pooled datasets or registries. Additionally, although corticosteroid use is typically more prevalent in non-ICI-treated cohorts due to antiemetic protocols (potentially biasing towards higher diabetes incidence in that group), our results

demonstrated the opposite trend, further emphasizing the robustness of our findings.

In summary, our propensity-matched analysis clearly illustrates a significantly increased risk of autoimmune diabetes and severe hyperglycemic events in patients treated with ICIs compared to those receiving non-ICI treatment. Specifically, we identified a 2.35-fold increased risk for T1DM and nearly a tenfold increased risk for diabetic ketoacidosis (DKA). These findings underscore the necessity of proactive monitoring, patient education, and multidisciplinary approaches to mitigate these serious adverse events. Further research into predictive biomarkers and underlying biological mechanisms is crucial as ICIs become increasingly integrated into oncology care.

Conclusions

ICIs have added a new dimension to cancer therapy and to the spectrum of iatrogenic autoimmune diseases. Our propensity-matched analysis confirms that treatment with immune checkpoint inhibitors significantly increases the hazard of developing a form of autoimmune diabetes and its most dangerous complication, DKA, in comparison to standard chemotherapy. Patients treated with ICIs had a 2.35-fold higher hazard of developing T1DM and, most concerning, a 10.58-fold higher hazard of developing life-threatening DKA.

These findings highlight the importance of active surveillance for hyperglycemia in patients on ICIs and prompt management to prevent catastrophic outcomes. The data also help identify higher-risk subgroups (such as those with underlying metabolic risk factors or on combination ICI regimens) in whom monitoring and patient education should be intensified. From a research perspective, our study contributes to a deeper understanding of the real-world impact of ICIs on endocrine function and underscores the need to unravel predictive biomarkers (e.g., HLA genotypes, T-cell profiles) for ICI-T1DM. As cancer survivorship improves, the long-term care of patients with ICI-induced diabetes will become an important aspect of survivorship planning. While ICIs continue to offer remarkable benefits in oncology, clinicians must remain vigilant for rare but severe immune-mediated adverse effects like T1DM, and be prepared to mitigate these risks through early detection and interdisciplinary care.

Abbreviations

ICIs	Immune checkpoint inhibitors
T1DM	Type 1 diabetes mellitus
DKA	Diabetic ketoacidosis
BMI	Body mass index
HbA1c	Hemoglobin A1c
irAEs	Immune-related adverse events
ICD	International Classification of Diseases

Supplementary Information

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Supplementary Material 1.

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Data availability

Protocol: Available from corresponding author upon reasonable request, Statistical Code: Available from corresponding author upon reasonable request. Data: de-identified Data is available at supplement.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board of Tri-Service General Hospital (TSGHIRB No.: E202516013). As the data used were de-identified and released for research purposes, informed consent from the participants was waived.

Competing interests

The authors declare no competing interests.

Declaration of generative AI in scientific writing

AI-assisted technology (ChatGPT by OpenAI) was used solely to refine language and formatting. The overall structure, content, and scientific interpretation were entirely determined by the authors.

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References

1. Brahmer J, Reckamp KL, Baas P, Crinò L, Eberhardt WE, Poddubskaya E, et al. Nivolumab versus docetaxel in advanced squamous-cell non-small-cell lung cancer. *N Engl J Med*. 2015;373:123–35.
2. Robert C, Long GV, Brady B, Dutriaux C, Maio M, Mortier L, et al. Nivolumab in previously untreated melanoma without BRAF mutation. *N Engl J Med*. 2015;372:320–30.
3. Gandhi L, Rodríguez-Abreu D, Gadgeel S, Esteban E, Felip E, De Angelis F, et al. Pembrolizumab plus chemotherapy in metastatic non-small-cell lung cancer. *N Engl J Med*. 2018;378:2078–92.
4. Hellmann MD, Paz-Ares L, Bernabe Caro R, Zurawski B, Kim S-W, Carcereny Costa E, et al. Nivolumab plus ipilimumab in advanced non-small-cell lung cancer. *N Engl J Med*. 2019;381:2020–31.
5. Motzer RJ, Escudier B, McDermott DF, George S, Hammers HJ, Srinivas S, et al. Nivolumab versus everolimus in advanced renal-cell carcinoma. *N Engl J Med*. 2015;373:1803–13.
6. Haanen J, Carbone F, Robert C, Kerr K, Peters S, Larkin J, et al. Management of toxicities from immunotherapy: ESMO clinical practice guidelines for diagnosis, treatment and follow-up. *Ann Oncol*. 2017;28:iv119–42.
7. Postow MA, Sidlow R, Hellmann MD. Immune-related adverse events associated with immune checkpoint blockade. *N Engl J Med*. 2018;378:158–68.
8. Barroso-Sousa R, Barry WT, Garrido-Castro AC, Hodi FS, Min L, Krop IE, et al. Incidence of endocrine dysfunction following the use of different immune checkpoint inhibitor regimens: a systematic review and meta-analysis. *JAMA Oncol*. 2018;4:173–82.
9. Byun DJ, Wolchok JD, Rosenberg LM, Giotra M. Cancer immunotherapy—immune checkpoint blockade and associated endocrinopathies. *Nat Rev Endocrinol*. 2017;13:195–207.
10. Chen X, Affinati AH, Lee Y, Turcu AF, Henry NL, Schiopu E, et al. Immune checkpoint inhibitors and risk of type 1 diabetes. *Diabetes Care*. 2022;45:1170–6.
11. Wright JJ, Salem J-E, Johnson DB, Lebrun-Vignes B, Stamatouli A, Thomas JW, et al. Increased reporting of immune checkpoint inhibitor-associated diabetes. *Diabetes Care*. 2018;41:e150–1.
12. De Filette J, Andreescu CE, Cools F, Bravenboer B, Velkeniers B. A systematic review and meta-analysis of endocrine-related adverse events associated with immune checkpoint inhibitors. *Horm Metab Res*. 2019;51:145–56.
13. Kotwal A, Haddox C, Block M, Kudva YC. Immune checkpoint inhibitors: an emerging cause of insulin-dependent diabetes. *BMJ Open Diabetes Res Care*. 2019;13;7(1):e000591. <https://doi.org/10.1136/bmjdr-2018-000591>
14. Clotman K, Janssens K, Specenier P, Weets I, De Block CE. Programmed cell death-1 inhibitor-induced type 1 diabetes mellitus. *J Clin Endocrinol Metab*. 2018;103:3144–54.
15. Marchand L, Thivolet A, Dalle S, Chikh K, Reffet S, Vouillarmet J, et al. Diabetes mellitus induced by PD-1 and PD-L1 inhibitors: description of pancreatic endocrine and exocrine phenotype. *Acta Diabetol*. 2019;56:441–8.
16. Godwin JL, Jaggi S, Sirisena I, Sharda P, Rao AD, Mehra R, et al. Nivolumab-induced autoimmune diabetes mellitus presenting as diabetic ketoacidosis in a patient with metastatic lung cancer. *J Immunother Cancer*. 2017;5:1–7.
17. Fife BT, Bluestone JA. Control of peripheral T-cell tolerance and autoimmunity via the CTLA-4 and PD-1 pathways. *Immunol Rev*. 2008;224:166–82.
18. Francisco LM, Sage PT, Sharpe AH. The PD-1 pathway in tolerance and autoimmunity. *Immunol Rev*. 2010;236:219–42.
19. Keir ME, Butte MJ, Freeman GJ, Sharpe AH. PD-1 and its ligands in tolerance and immunity. *Annu Rev Immunol*. 2008;26:677–704.
20. Sharpe AH, Pauken KE. The diverse functions of the PD1 inhibitory pathway. *Nat Rev Immunol*. 2018;18:153–67.
21. Wright JJ, Powers AC, Johnson DB. Endocrine toxicities of immune checkpoint inhibitors. *Nat Rev Endocrinol*. 2021;17:389–99.
22. Stamatouli AM, Quandt Z, Perdigoto AL, Clark PL, Kluger H, Weiss SA, et al. Collateral damage: insulin-dependent diabetes induced with checkpoint inhibitors. *Diabetes*. 2018;67:1471–80.
23. Quandt Z, Young A, Anderson M. Immune checkpoint inhibitor diabetes mellitus: a novel form of autoimmune diabetes. *Clin Exp Immunol*. 2020;200:131–40.
24. Gauci M-L, Laly P, Vidal-Trecan T, Baroudjian B, Gottlieb J, Madjlessi-Ezra N, et al. Autoimmune diabetes induced by PD-1 inhibitor—retrospective analysis and pathogenesis: a case report and literature review. *Cancer Immunol Immunother*. 2017;66:1399–410.
25. Takada S, Hirokazu H, Yamagishi K, Hideki S, Masayuki E. Predictors of the onset of type 1 diabetes obtained from real-world data analysis in cancer patients treated with immune checkpoint inhibitors. *Asian Pac J Cancer Prevention: APJCP*. 2020;21:1697.

26. Zhong L, Coe SP, Stromberg AJ, Khattar NH, Jett JR, Hirschowitz EA. Profiling tumor-associated antibodies for early detection of non-small cell lung cancer. *J Thorac Oncol.* 2006;1:513–9.
27. Noble JA, Valdes AM. Genetics of the HLA region in the prediction of type 1 diabetes. *Curr Diab Rep.* 2011;11:533–42.

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